

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_104046 | Base: main-v1.0

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Decision: GO_WITH_CONSTRAINTS | Confidence: 78%

CHANGE REQUEST

togliamo la telecamera

CASCADE ISSUES IDENTIFIED

[CRITICAL] ISSUE-001 — Loss of optical resolution requirement compliance

Subsystem: optical | Req: L1-OPT-001 | Type: constraint_violation

Removing the telescope tube will eliminate the primary imaging capability, violating the $GSD \leq 1.5m$ at nadir requirement.

[CRITICAL] ISSUE-002 — Thermal control system failure

Subsystem: thermal | Req: L1-THM-001 | Type: constraint_violation

The thermal panel is removed, which will cause the optical bench temperature to exceed allowable ΔT limits across orbital cycles.

[MAJOR] ISSUE-003 — Structural integrity compromised

Subsystem: mechanical | Req: None | Type: reanalysis_required

The removal of the optical subsystem will alter the structural balance and potentially reduce the first natural frequency below the required 80 Hz.

[MAJOR] ISSUE-004 — Star tracker alignment issues

Subsystem: startracker | Req: None | Type: reanalysis_required

Without the optical subsystem, the star tracker may lose its reference point for accurate pointing and attitude determination.

[MINOR] ISSUE-005 — Data processing pipeline invalidation

Subsystem: dataproc | Req: None | Type: reanalysis_required

The data processing algorithms will no longer have valid input from the optical subsystem, rendering them obsolete.

MISSING ECSS TESTS — NOW REQUIRED

No additional tests required.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	61%	WARN	ΔT 14.8°C — margin < 5%	—
mechanical	88%	OK	1st mode 112 Hz > 80 Hz req	—
structure	91%	OK	Mass 33.8 kg / budget 35 kg	—
solar	89%	OK	Power margin +18%	—
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

No requirements directly invalidated.